

Clear Understanding of the Problem

Virtual Try-On (VTON) systems aim to digitally dress a person with a garment using only images. The goal is to generate a realistic image of a person wearing a selected clothing item without physically trying it on. This technology has significant importance in the fashion and e-commerce industry because it helps customers visualize how clothes may look on them before purchasing, thereby improving the online shopping experience and potentially reducing product returns.

The fundamental task of a virtual try-on system is to take two inputs: an image of a person and an image of a target garment, and generate a new image in which the person appears to be wearing that garment. While the idea appears simple, achieving high-quality results is technically challenging.

One of the primary difficulties is **garment alignment**. Clothes in product images are usually captured in a flat or mannequin pose, while the person in the target image may have a completely different pose or body shape. The system must therefore accurately transform and align the garment so that it fits the person's body naturally.

Another major challenge is **preserving garment details**. Clothing often contains fine textures, patterns, logos, and folds that must remain visible after transformation. If these details are not preserved, the generated image may look unrealistic or lose important product characteristics.

Occlusion handling is another important problem. Parts of the body such as arms or hands may cover sections of the clothing, and the system must correctly determine which regions of the garment should be visible and which should be hidden.

Additionally, the generated image must maintain the **identity and pose of the person**. The system should not distort the person's body structure or change their appearance while integrating the new clothing.

Recent research has addressed these challenges using deep learning approaches such as Generative Adversarial Networks (GANs), flow-based warping models, and diffusion-based generative models. However, existing methods still struggle to balance **high visual realism, accurate clothing alignment, and computational efficiency**. Many models produce visually appealing results but require large computational resources, making them difficult to deploy in real-world applications.

Therefore, the core problem addressed in this project is the development of a virtual try-on system that can generate **high-quality, realistic images while accurately preserving garment structure and maintaining efficient computation**.

Proposed Solution

To address the challenges of virtual try-on systems, this project proposes a hybrid deep learning framework that combines garment alignment techniques with modern image generation models to produce realistic and high-quality results.

The first stage of the system focuses on **person representation and preprocessing**. The input image of the person is processed to extract important structural information such as body pose and semantic regions (e.g., arms, torso, and head). This representation helps the model understand the spatial structure of the human body while removing the original clothing information so that the new garment can be applied effectively.

The second stage involves **garment alignment and transformation**. Since the clothing image may not match the pose of the person, a warping mechanism is used to reshape and align the garment according to the body structure of the person. This step ensures that sleeves, collars, and other clothing components are positioned correctly and follow the natural shape of the body.

Once the garment is aligned, the system proceeds to the **image synthesis stage**, where a generative model produces the final image of the person wearing the new clothing. A diffusion-based or deep generative model is used in this stage to reconstruct the image while maintaining high visual realism. The model integrates information from both the person representation and the warped garment to generate a coherent and natural-looking result.

To further enhance visual quality, a **refinement module** is applied after image generation. This module corrects artifacts, improves texture consistency, and ensures smooth blending between the garment and the person's body. As a result, the final output image appears more realistic and visually appealing.

The system is trained using established virtual try-on datasets such as **VITON or VITON-HD**, which contain paired images of people and garments. The performance of the model will be evaluated using image quality metrics such as **Structural Similarity Index (SSIM)**, **Fréchet Inception Distance (FID)**, and **Learned Perceptual Image Patch Similarity (LPIPS)**, along with visual comparisons to baseline models.

By combining accurate garment warping with powerful generative modeling and refinement techniques, the proposed system aims to produce realistic virtual try-on results while maintaining efficiency and preserving fine garment details. This approach can contribute to more practical and reliable virtual try-on applications in online fashion platforms.